

RETURN RISK ANALYTICS SYSTEM

Final Documentation Report

Project Report Submitted for Internship Project Submission

Submitted By

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Internship Role: Intern (AI/ML)

Project Title: Return Risk Analytics System

Technology Stack:

Python | Machine Learning | Classification | Scikit-Learn | XGBoost | Streamlit | Plotly | Pandas

Project Description:

A machine learning classification system developed to predict product return probability and identify high-risk orders. The project includes exploratory data analysis, feature engineering, model comparison, hyperparameter tuning, risk scoring, high-risk order identification, and an interactive Streamlit dashboard for business decision support.

Submitted To

Company/Organization Name: Data Vidwan

Mentor/Manager Name: Meet Mistry

Date of Submission:

16/06/2026

RETURN RISK ANALYTICS SYSTEM

1. Project Objective

The objective of this project is to build a machine learning classification model that predicts whether an e-commerce order is likely to be returned. The system generates return risk scores, identifies high-risk orders, and provides business insights through a professional Streamlit dashboard.

2. Requirement Coverage

Requirement	Status	Project File / Output
Product Return Prediction Model	Completed	models/return_prediction_model.pkl
Feature Schema	Completed	models/model_features.pkl
Processed Dataset	Completed	data/processed/final_scored_dataset.csv
EDA Report	Completed	notebooks/day1.ipynb
Classification Evaluation	Completed	notebooks/day2.ipynb
Risk Scoring System	Completed	final_scored_dataset.csv
High Risk Order Report	Completed	reports/High_Risk_Orders.csv
Return Analysis Dashboard	Completed	Dashboard/app.py
Project Documentation	Completed	Documentation_Report.docx

3. Dataset

Dataset: Product Return Prediction Dataset

Raw Records: 1,500

Processed Records: 1,500

Returned Orders: 406

Non-Returned Orders: 1,094

Return Rate: 27.1%

Features: order_id, product_id, price, rating, return_status.

4. Work Completed

- Loaded and inspected the raw dataset.
- Performed missing value and data quality validation.
- Conducted exploratory data analysis and generated visualizations.
- Created price_bucket feature.

- Created low_rating_flag feature.
- Created product_order_count feature.
- Trained Logistic Regression, Random Forest, and XGBoost models.
- Compared model performance using classification metrics.
- Performed hyperparameter tuning using GridSearchCV.
- Generated return probability and risk scores.
- Identified high-risk orders.
- Developed a professional Streamlit dashboard.
- Exported trained model and feature schema.

5. Methodology

5.1 Data Validation

No missing values or major inconsistencies were identified.

5.2 Exploratory Data Analysis

Return distribution, rating distribution, price analysis, and correlation analysis were performed.

5.3 Feature Engineering

price_bucket, low_rating_flag, and product_order_count were engineered.

5.4 Model Development

Logistic Regression, Random Forest, and XGBoost were evaluated.

5.5 Hyperparameter Tuning

GridSearchCV selected C=0.1 and solver=liblinear for Logistic Regression.

6. Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7133	0.4510	0.2840	0.3485	0.7111
Random Forest	0.6300	0.3026	0.2840	0.2930	0.5772
XGBoost	0.6967	0.3958	0.2346	0.2946	0.6482
Tuned Logistic Regression	0.7233	0.4783	0.2716	0.3465	0.7033

7. Selected Model

- Tuned Logistic Regression Classifier

After evaluating multiple machine learning algorithms, including Logistic Regression, Random Forest, and XGBoost, the **Tuned Logistic Regression Classifier** was selected as the final model for deployment.

- Reason for Selection

The model achieved the strongest overall performance among the evaluated models and demonstrated the best balance between predictive capability and interpretability.

Final Model Performance:

8.	Metric	Score
	Accuracy	72.33%
	Precision	47.83%
	Recall	27.16%
	F1 Score	34.65%
	ROC-AUC	70.33%

Streamlit Application

The Streamlit application is implemented in **Dashboard/app.py**. It includes the following pages:

- Executive Overview
- EDA Insights
- Prediction Center
- High-Risk Orders

The Executive **Overview** page presents key business metrics, return distribution, risk category summaries, model performance metrics, and overall return analytics. The EDA Insights page displays exploratory data analysis visualizations including return trends, rating distribution, price analysis, correlation analysis, and feature importance. The Prediction Center page allows users to enter order details and generate real-time return probability predictions along with risk categorization. The High-Risk Orders page provides an interactive view of identified high-risk orders, risk scores, filtering capabilities, and downloadable reports for business monitoring and decision-making.

9. Key Findings

- Return Rate: 27.1%
- Lower ratings significantly increase return probability.
- Rating emerged as a major return driver.

- 59 high-risk orders were identified.
- Risk scoring enables proactive business intervention.

10. Recommendations

- Monitor low-rated products.
- investigate recurring product issues.
- prioritize quality checks for high-risk orders.

11. Achievements

- Completed an end-to-end machine learning workflow including EDA,
- feature engineering,
- model development,
- risk scoring,
- dashboard deployment,
- business insight generation.

12. Limitations

The dataset contains limited transactional attributes and lacks customer demographic information and return-reason details.

13. Final Submission Contents

- Documentation Report,
- Model Files,
- Streamlit Application Files(dashboard/)
- Processed Dataset(data/)
- High-Risk Orders Report
- Notebooks
- README
- requirements.txt.

14. How to Run

```
pip install -r requirements.txt  
streamlit run Dashboard/app.py
```

15. Final Conclusion

The Return Risk Analytics System successfully delivers an end-to-end machine learning solution for return prediction, risk scoring, and high-risk order identification. The deployed

dashboard provides actionable insights for business decision-making and return reduction strategies.